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Choroidal Melanoma Masquerading as Central Serous Chorioretinopathy

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Purpose: To determine the clinical features and outcomes of choroidal melanoma initially masquerading as central serous chorioretinopathy (CSCR).
Design: Retrospective case series.
Subjects: All patients with choroidal melanoma, initially misdiagnosed as CSCR elsewhere and evaluated by the Ocular Oncology Service at Wills Eye Hospital from 2004 to 2022, were included.
Methods: A retrospective detailed review of patient charts and imaging was performed for all patients included in the study. Paired *t* tests and chi-squared tests were performed for data analysis.
Main Outcome Measures: The primary outcome measures included clinical characteristics, ultrasonography, OCT, fundus autofluorescence, fluorescein angiography, and indocyanine green angiography. The secondary outcome measures included treatment results, such as the final visual acuity, tumor control, radiation-related complications, and melanoma-related metastlasis and death.
Results: There were 22 patients (mean age, 48 years; 16 men) in this cohort. The mean interval between initial CSCR diagnosis and suspicion of choroidal melanoma was 50 months (median, 50 months; range, 0–242 months). At tumor diagnosis, the melanoma was submacular in 16 (73%) patients and extramacular in 6 (27%) patients. The mean tumor thickness was 3.4 mm (median, 2.5 mm; range, 1.4–10.7 mm), and the mean basal diameter was 9.2 mm (median, 8.0 mm, range, 4.5–22.0 mm). Features enabling differentiation of choroidal melanoma from CSCR (affected versus unaffected eye) included choroidal thickness asymmetry (100% $>$ 300 μm versus 21% $>$ 300 μm; $P = 0.005$), ipsilateral choroidal surface irregularity (100% versus 0%; $P < 0.001$), loss of choroidal vascular detail on OCT (100% versus 0%; $P < 0.001$), presence of multiple pinpoint leaks on angiography (100% versus 0%; $P < 0.001$), and contralateral lack of autofluoresence abnormalities (75% versus 6%; $P = 0.001$). Management of the choroidal melanoma included plaque radiotherapy (19, 86%), enucleation (2, 9%), or treatment elsewhere (1, 5%). On follow-up (mean, 6 years), vision loss of $\geq$ 3 Snellen lines (9 patients, 47%), metastasis (3 patients, 14%), and death (1 patient, 5%) were noted.
Conclusions: Patients with presumed CSCR, especially if chronic, should be evaluated for a possible thin underlying choroidal melanoma with a dilated fundus examination and multimodal imaging. *Ophthalmology Retina 2023;7:171-177 © 2022 by the American Academy of Ophthalmology*

Choroidal melanoma is an uncommon malignancy with a mean age-adjusted incidence of 5 per million in the United States.[1] Given its rarity, practitioners may not have it at the forefront in their differential diagnosis of conditions that produce subretinal fluid (SRF), especially when a young patient presents with such findings.

Ophthalmic conditions that are often mistakenly diagnosed as choroidal melanoma (pseudomelanomas) have been well documented, including choroidal nevus (49%), peripheral exudative hemorrhagic chorioretinopathy (6%), congenital hypertrophy of the retinal pigment epithelium (6%), idiopathic hemorrhagic detachment of the retina or pigment epithelium (5%), and circumscribed choroidal hemangioma (5%).[2,3] On the contrary, choroidal melanoma can simulate other ophthalmic conditions, including choroidal nevus, metastasis, hemangioma, osteoma, focal scleral nodule, sclerochoroidal calcification, posterior scleritis, and others. There are reports of unilateral glaucoma with angle pigmentation from underlying angle melanoma[4–7] and presumed orbital cellulitis, chronic uveitis, endophthalmitis, and choroidal hemorrhage that eventually proved to represent choroidal melanoma.[8–12]

Choroidal melanoma masquerading as central serous chorioretinopathy (CSCR) has only been reported in a single case report.[13] The reported mean age-adjusted incidence of CSCR is 10 per 100 000 males and 1.7 per 100 000 females.[14] In this study, we analyzed a consecutive case series of choroidal melanomas that were initially presumed to represent CSCR. Our goal was to identify key features on multimodal imaging that distinguish between CSCR and choroidal melanoma and raise awareness that choroidal malignancies can be important mimickers of CSCR.

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